



Figure 1: HTML page

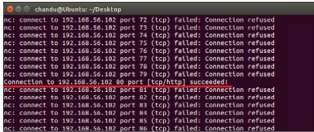


Figure 2: Result of port scanning

listener and <PORT> is the port of the listener.

Here is an example.

```
chandu@ubuntu:~$ nc -lp 1234
Hello OSFY readers !!!
```

```
chandu@ubuntu:~$ nc 192.168.56.103 1234
Hello OSFY readers !!!
```

The server stops if either of the users stops chatting. To make the chat server permanent, you can add a -k tag in the listener terminal.

An example is given below.

```
Listener : nc -k -lp 1234
```

Web server: A simple static Web page can be hosted on a specific port on a system using the following commands:

```
nc -lp <PORT> -q 1 < index.html
```

...where *index.html* is an HTML page on the hosted system as below.

```
chandu@ubuntu:~$ cat index.html
<html>
<head>
<title>Hello</title>
</head>
<body>
<h2> Hello OSFY readers !</h2>
</body>
</html>
```

If you wish to open the Web page using a browser, you could set the port to be 80.

```
nc -lp 80 -q 1< index.html
```

When you open a browser and enter the IP address of the computer as the URL, you will get the contents of your HTML page [Figure 1]. When you see the terminal, you will find a lot of text, which is actually implemented at the background when browsing. But if you reload the page, you cannot find the page, because Netcat stops successfully after sending the contents of the HTML file.

To make a permanent Web server on a port 80, use a *while* loop:

```
while true; do nc -l 8888 -q 1 < index.html; done
```

Note: You should do this as a root user because starting a service requires root privileges.

Port scanning: Port scanning is very important for sys admins and pentesters. To use Netcat for port scanning use the following command:

```
nc -V -z -w <SEC> <IP ADDRESS> < PORT RANGE: 1-1000>
```

...where,
-v: verbose mode
-w: wait for
SEC: number of seconds to wait
-z: operate in zero IO mode, i.e., display without information and without analysing the response.

For example:

```
nc -V -z -w 1 192.168.56.102 1-100
```

You could scroll up to view which ports are open [Figure 2]. You could also scan one port at a time using the exact port.

```
nc -V -z -w 1 192.168.56.102 80
```

By default, Netcat uses a TCP connection. You can also scan domain names in a UDP connection. You have to use -u, as follows:

```
nc -V -z -w <SEC> -u <IP ADDRESS> <PORT RANGE: 1-1000>
```

...where -u is the UDP connection.

Note: If the Netcat-traditional version is used for port scanning, '-v' should be replaced by '-vv'. The output will show only the open ports. It hides the 'Connection refused' information, thus making the output easy to read.

File transfer: Using the same logic of the chat server, one could send and receive the contents of a file. You could do this in two ways.

1. If you want to send a single file, you could directly send it using the following commands.